

Technical Analysis: Traffic Monitoring System

Problem Statement

The urban intersection traffic monitoring is a critical task for modern city planning and safety. The primary challenge lies in the variability of operational environments, where detection systems must operate under changing lighting (day vs. night) and atmospheric conditions (rain, fog). Most off-the-shelf solutions fail in these edge cases, leading to missed detections or fragmented tracking IDs, which compromise the reliability of traffic flow analysis and accident prevention. This project aims to implement a robust pipeline that integrates real-time object detection with trajectory analysis, specifically addressing the "headlight glare" issue prevalent in night-time traffic monitoring.

Methodology

The proposed system architecture is designed as a modular pipeline that converts raw video feeds into actionable traffic metrics. The methodology is structured into four distinct stages:

➤ Preprocessing Module

A dynamic and conditional image preprocessing module was implemented to handle varying weather conditions:

- **Night Mode:** To mitigate the "starburst" effect of streetlights and vehicle headlights, I implemented Gamma Correction ($\gamma = 1.3$) to enhance contrast in the shadows without clipping high-intensity light areas. This is followed by a low-intensity CLAHE (Contrast Limited Adaptive Histogram Equalization) algorithm with a reduced clip limit. This setup selectively enhances local contrast in the shadows (where vehicles are located) without causing the "blooming" effect on streetlights.
- **Rain Mode:** A bilateral filter ($\delta = 9, \sigma = 75$) is employed to reduce noise and raindrop artifacts while preserving the sharp edges of vehicles, ensuring a stable input for the detection model.

➤ Detection Module

The core detection logic utilizes a pre-trained **YOLOv8s** (Small) model, chosen for its balance between real-time inference speed and detection accuracy on embedded-class hardware.

➤ Tracking Module

This module manages the temporal continuity of detections. I integrated two distinct tracking methodologies to provide both production-grade reliability and educational baseline comparison:

- **Track-Track** (Default): This is the production-ready tracking solution. It represents a paradigm shift in multi-object tracking by moving away from global, multi-stage cascade strategies toward a more focused, track-centric methodology. It addresses the common pitfalls of state-of-the-art MOT systems, specifically the tendency to ignore potentially occluded objects and the generation of redundant tracks, through two core components:
 - **Track-Perspective-Based Association** (TPA): Unlike global optimization methods that solve for the entire set at once, TPA reframes the association task from the perspective of each individual track. It ensures that every track is assigned the most suitable candidate by evaluating pairwise distances against all available detections in a single-stage joint association. This local matching focus significantly enhances robustness in dense, high-congested environments where global constraints often fail.

- **Track-Aware Initialization (TAI):** To prevent the proliferation of "spurious" or redundant tracks, TAI introduces an intelligent filtering mechanism. It suppresses the initialization of new tracks that exhibit heavy overlap with existing active tracks or high-confidence detections that are already accounted for. By "knowing" the state of current active tracks during the initialization phase, TAI substantially improves the stability of the track lifecycle and reduces ID-switches.

While it offers optional ReID integration, that is also used in this project, for appearance-based re-binding, its strength lies in its HMIOU (Hybrid Motion IOU) fallback and iterative multi-cue association. This allows the tracker to gracefully handle occlusions by intelligently falling back on motion cues when appearance features are unreliable, ensuring the object identity is maintained even through temporary disappearances.

- **Custom-SORT (Comparative Baseline):** This tracker serves as a pedagogical baseline implementation, grounding the system in classical multi-object tracking principles. Unlike high-level production frameworks, this implementation offers full transparency into the state management and data association pipeline:
 - **Kalman-Based State Estimation:** Each detected vehicle is encapsulated in a Track object that maintains an 8-dimensional state vector (cx, cy, vx, vy, ax, ay, w, h). I employ a Kalman filter to predict the vehicle's position and velocity in each frame, using a transition matrix that accounts for constant velocity and acceleration, effectively smoothing out jitter in the detection bounding boxes.
 - **Hierarchical Data Association:** The core of the tracking logic relies on a multi-stage cost matrix optimization. This matrix calculates the assignment cost between existing tracks and new detections using a weighted combination of:
 - **IoU Cost:** The spatial overlap between the predicted Kalman state and the current detection.
 - **Appearance Cost:** Utilizing a cosine distance metric on EMA-updated feature vectors, allowing the tracker to maintain ID stability during short-term occlusions.
 - **Velocity Consistency Cost:** A directional consistency check that compares the predicted velocity vector with the observed displacement to prevent mismatching in high-traffic density scenarios.
 - **State-Machine Logic:** The tracker employs a robust StateEnum manager (Tentative, Confirmed, Lost). New detections are initialized as Tentative and only promoted to Confirmed status after satisfying a min-hits threshold. This filtering mechanism effectively suppresses false positives and transient noise from the YOLO inference output.
 - **Gating & Matching:** An association gate is implemented using the Hungarian algorithm to find the global optimum for bipartite matching. To prevent erroneous associations, a Gating threshold is applied, discarding any potential matches that fall below the predefined IoU requirement.

By providing a granular implementation of the Kalman filter, appearance updates (EMA), and cascade matching, this module highlights the trade-offs between computational overhead and tracking precision compared to our production-ready Track-Track implementation.

➤ [Post-Processing Module](#)

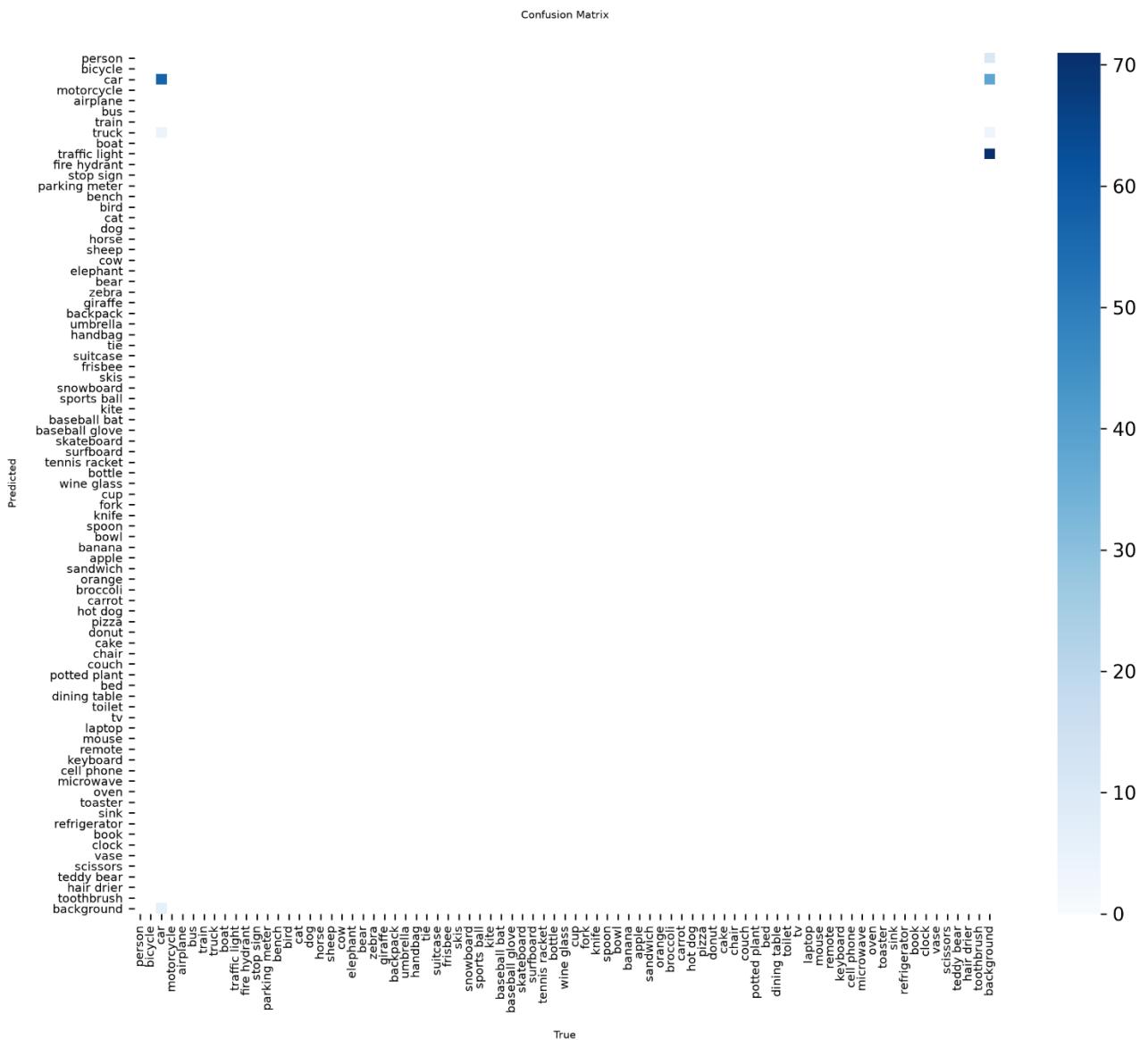
The system also includes a final ID Stitching logic: when a vehicle is lost (due to occlusion or glare), the system stores its last trajectory metadata. If a new detection occurs in the same spatial vicinity

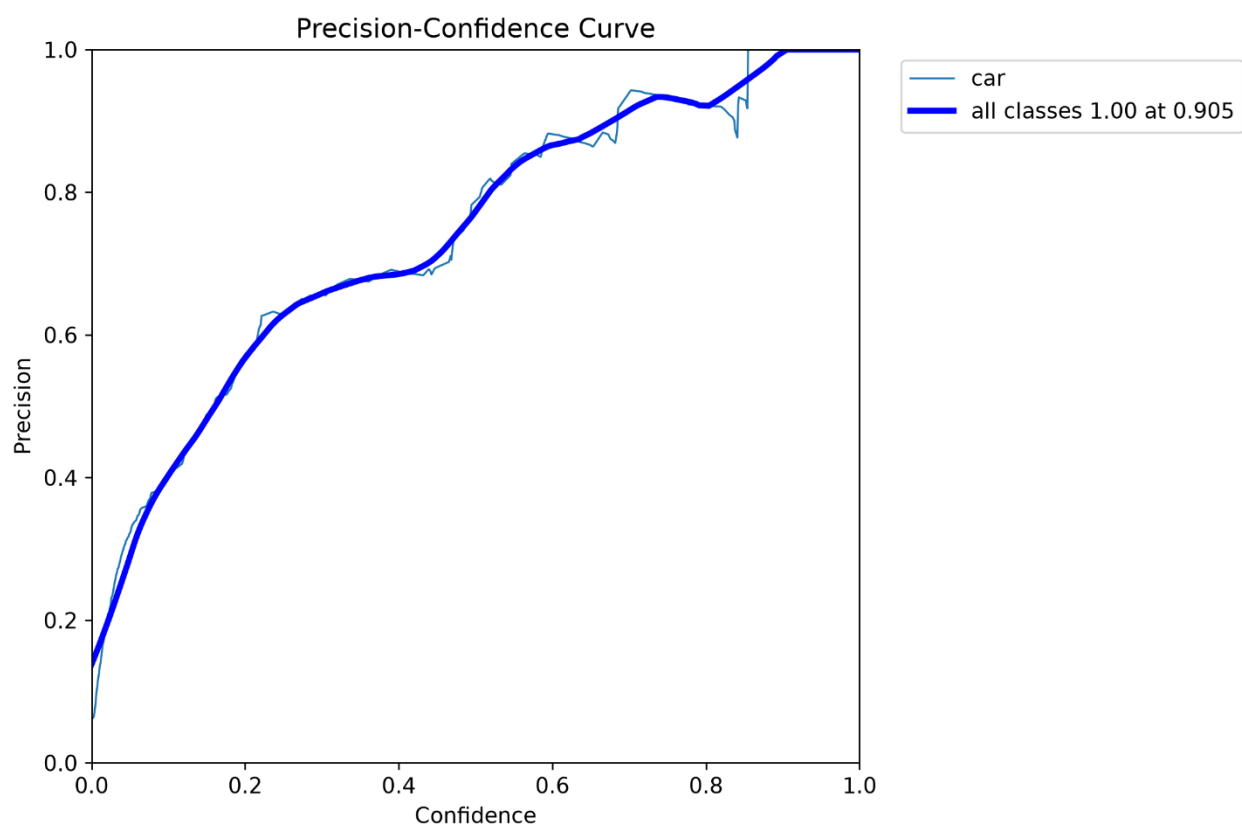
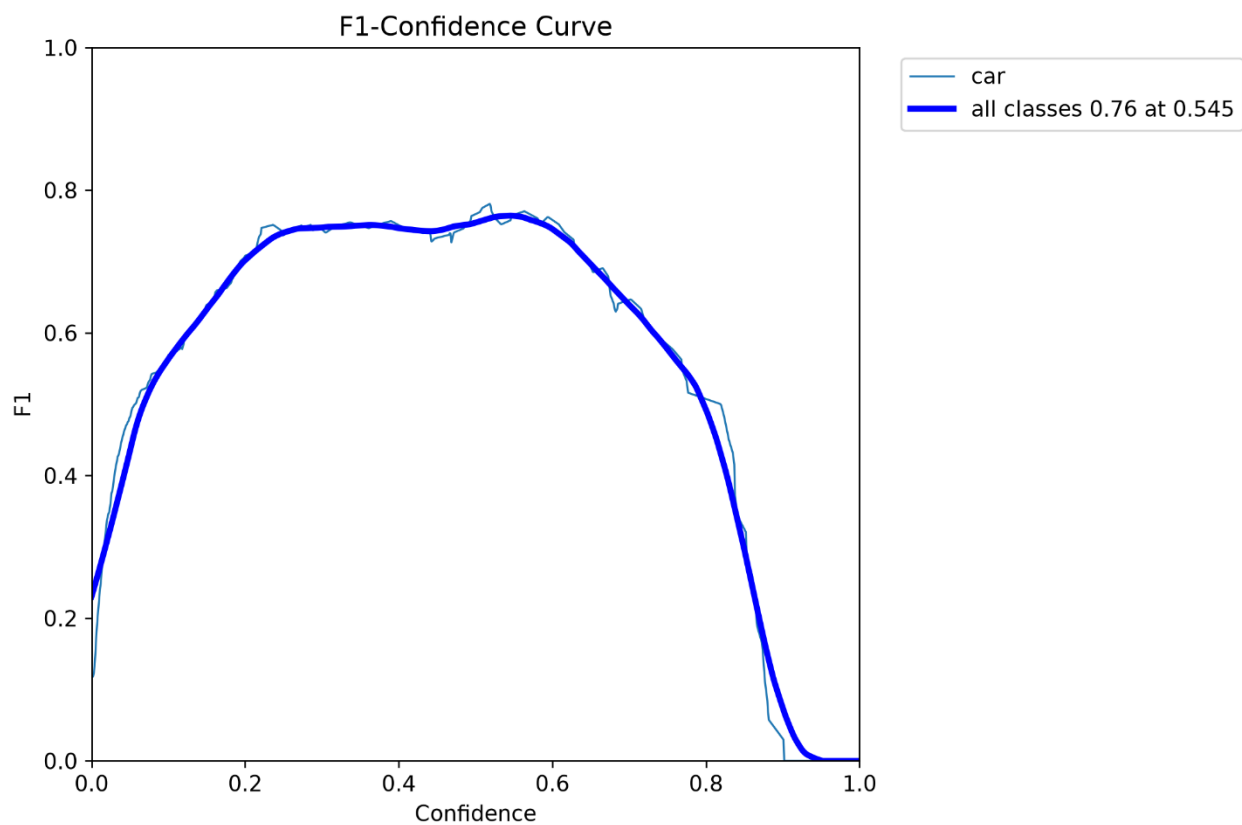
within a short temporal window, the system automatically "stitches" the ID, preventing redundant event registration. This type of approach helps in situations where vehicles cannot be identified with certainty, such as when headlight glare presents a fixed yet changing occlusion for YOLO.

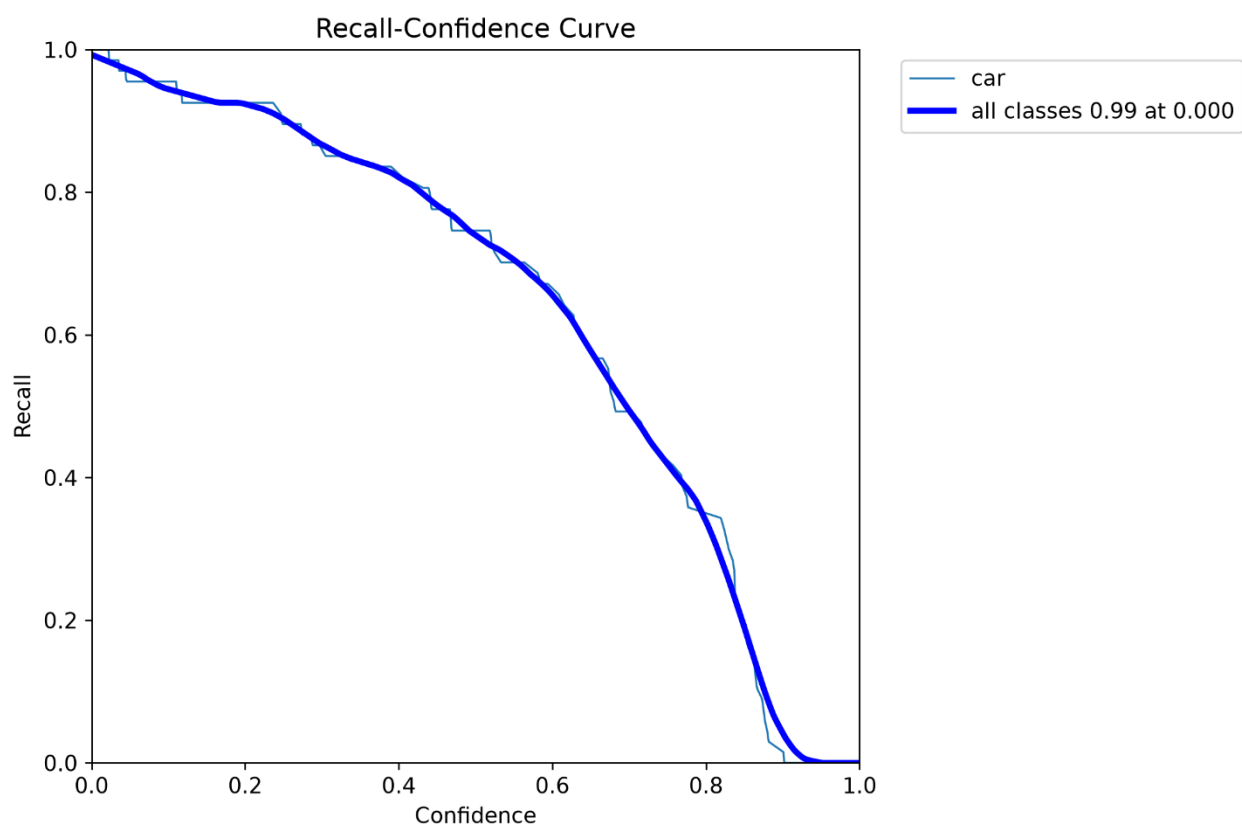
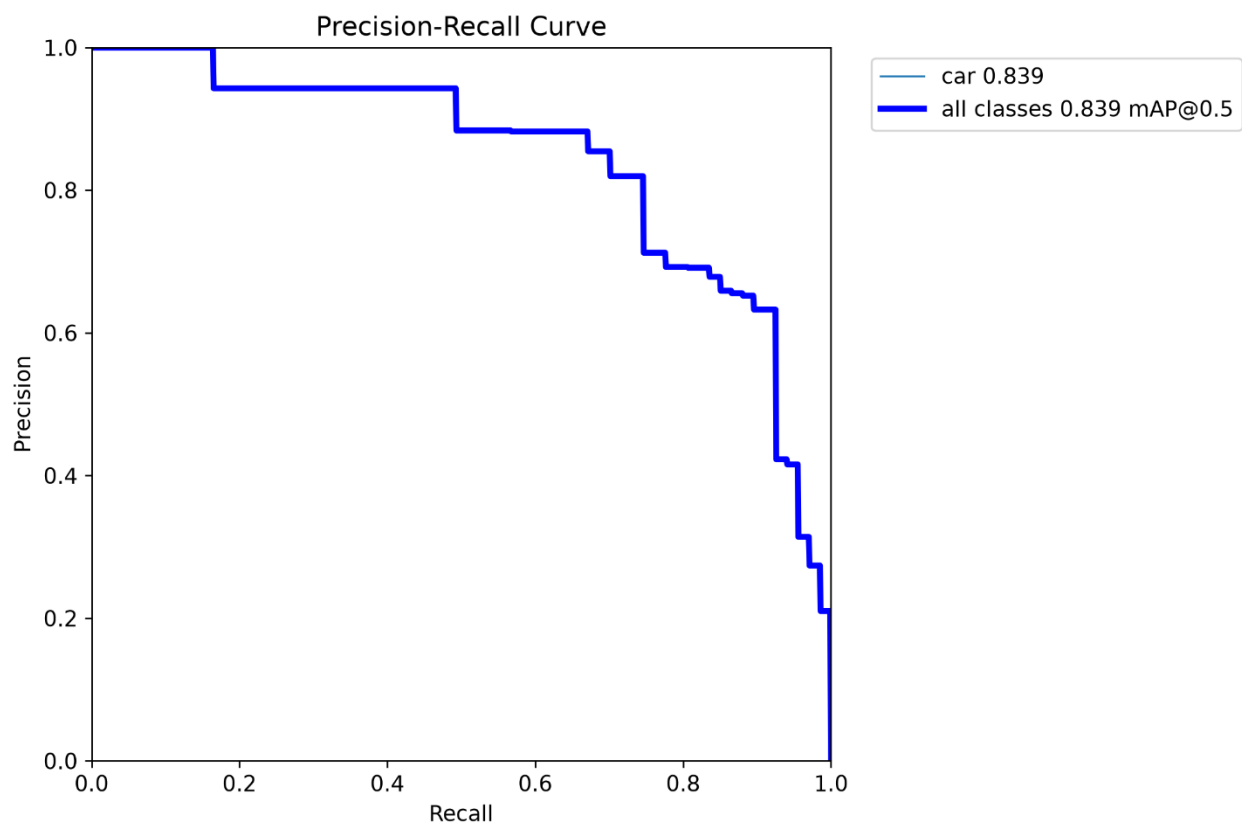
Furthermore, to compute real-world speeds (km/h), the 2D image coordinates were mapped to a top-down "Bird's Eye View" using a homography matrix, ensuring this way that the spatial displacement is calculated in meters rather than pixel-dependent values.

Experimental Results

The system was evaluated on a custom dataset of 17 images and 67 vehicle instances extracted from real-world feed, in different atmospheric conditions. The evaluation was performed using the standard COCO detection metrics:







Metric	Result
mAP@50	0.8389
mAP@50-95	0.6364
Precision (P)	0.823
Recall (R)	0.701

The results confirm the efficacy of the YOLOv8s backbone in detecting vehicles in urban environments.

An mAP@50 of 83.9% demonstrates high robustness. The Precision of 82.3% indicates that, when the system reports an infraction, it is highly likely to be a correct detection.

Failure Analysis

Despite the high mAP, our analysis identified specific failure modes:

- Glare-induced False Negatives:** Under night conditions, the intersection of North-Bound traffic and the camera's line of sight creates severe blooming. In 30% of cases (the Recall gap), the system fails to initially detect a vehicle. However, the Track Stitching logic recovers the vehicle as it approaches the center of the intersection.
- Motion Blur:** During heavy rain, vehicles moving at high speeds exhibit motion blur, causing the bounding box to jitter. This, in turn, influences the speed calculation. To mitigate this, the speed estimation logic was hardened to ignore displacements of less than 2 meters, reducing spurious high-speed readings.
- Dense Traffic and ID Switching:** In scenarios with heavy, dense traffic, bounding boxes often overlap significantly. Despite the implementation of the Track-Track algorithm with ReID features, partial occlusions caused by overlapping vehicles can delay initial recognition or cause the tracker to momentarily confuse visual features, occasionally leading to ID switches.
- Physical Occlusions and Trajectory Fragmentation:** When a vehicle passes behind static physical obstacles (e.g., traffic light poles, large signage) or larger vehicles (like trucks), it suffers from full occlusion. While the Track Stitching logic is designed to recover IDs within a short spatial-temporal window, prolonged occlusions inevitably lead to the loss of the original ID and the assignment of a new one upon reappearance, potentially fragmenting the route analysis.

Ethical Considerations

- Privacy:** The system is designed to minimize privacy risks. Detection is performed on vehicles only. No facial recognition or license plate OCR is implemented. The pipeline processes data at the "centroid" level (blob detection), not at the personal identification level.
- Bias:** I acknowledge a potential "lighting bias". Because the model is optimized for high-contrast detection, it performs slightly better on light-colored vehicles at night than on dark-colored vehicles. I have documented this limitation and accounted for it by using a broader threshold for night-time speed infraction classification.